

House Price Prediction in King County, Washington

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Abstract. We analyse the King County House Sales dataset (21,612 residential transactions, 2014–2015) using a complete machine learning pipeline implemented in KNIME. Four regression models are compared: Linear Regression (LR), Decision Tree (DT), Random Forest (RF), and Gradient Boosted Trees (GBT). RF achieves the best test performance ($R^2=0.703$, $RMSE=0.292$), confirmed by 5-fold cross-validation. PCA with 12 components (95% variance) yields $R^2=0.654$ via LR on reduced features. K-Means clustering with $K=2$ (Silhouette=0.248) identifies two distinct market segments. Living space, construction grade, and bathroom count are the primary price determinants.

Introduction

Accurate residential property valuation is important for buyers, sellers, and financial institutions. Machine learning offers a data-driven alternative to traditional appraisal methods. This study addresses the question: *which structural and locational features most strongly determine house prices in King County, and how accurately can ML models predict them?*

We compare four regression algorithms, reduce dimensionality with PCA, validate all models via 5-fold cross-validation, and segment the market with K-Means clustering. Three alternative approaches exist for this problem: hedonic regression, spatial econometrics, and deep learning. Our approach differs by combining model comparison, dimensionality reduction, and unsupervised market segmentation in a single reproducible KNIME workflow. Our main contributions are: (i) a rigorous model comparison on a standardised pipeline; (ii) PCA analysis quantifying the cost of compression; (iii) geographic market segmentation.

Data and Preprocessing

The raw dataset [1] contains 21,613 rows and 21 columns. The target is `price` (\$75k–\$7.7M). One outlier (33 bedrooms, 1,620 sqft) is removed, leaving 21,612 records.

Feature engineering. Three derived features replace raw columns: `house_age` = `2015 - yr_built`; binary `renovated` (1 if `yr_renovated` > 0); binary `basement` (1 if `sqft_basement` > 0). Columns `id`, `date`, `zipcode`, `yr_built`, `yr_renovated`, `sqft_basement`, and `sqft_above` are removed as non-informative or redundant. The target is log-transformed (`log_price` = $\ln(\text{price})$) to normalise the right-skewed distribution (Figure 1). This is motivated by the fact that price relationships are multiplicative, not additive. Z-score normalisation is fitted on the training set only and applied to the test set via a separate Normalizer Apply node, preventing data leakage. The final split is 70/30 (15,128 train, 6,484 test, seed=42), yielding 14 features: `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, `waterfront`, `view`, `condition`, `grade`, `sqft_living15`, `sqft_lot15`, `house_age`, `renovated`, `basement`.

Correlation structure. Table 1 reports the strongest pairwise Pearson correlations. `sqft_living` and `grade` are the features most correlated with `log_price`. High mutual correlation among size features (`sqft_living`, `sqft_living15`, `bathrooms`) motivated the PCA analysis in Section 3.

Table 1: Selected pairwise correlations ($|\rho| > 0.44$). All are statistically significant.

Feature pair	ρ	Note
<code>sqft_living</code> – <code>grade</code>	0.763	Size and quality linked
<code>sqft_living</code> – <code>sqft_living15</code>	0.756	Neighbourhood size effect
<code>sqft_living</code> – <code>bathrooms</code>	0.755	Larger homes, more amenities
<code>grade</code> – <code>bathrooms</code>	0.665	Quality and amenity count
<code>floors</code> – <code>house_age</code>	−0.489	Newer homes multi-floor
<code>grade</code> – <code>house_age</code>	−0.447	Newer homes higher grade

Methods

Regression models. We train four models to predict `log_price`. LR provides an interpretable baseline. DT (max depth=8, min 5 records/node) captures non-linearities. RF (100 trees, seed=42) and GBT (200 trees, lr=0.05, depth=6, row sampling=0.8) are ensemble methods expected to outperform single models.

PCA. PCA Compute is applied to the 14 normalised training features (`price` and `log_price` excluded). The number of components is chosen to retain 95% of total variance, yielding 12 principal components. We then train a new LR on the projected training set and evaluate on the projected test set. This allows us to quantify precisely how much predictive information is lost by compressing 14 features into 12 PCs.

Cross-validation. Independent 5-fold CV loops (X-Partitioner / X-Aggregator, seed=42, linear sampling) are run for each of the four models on the full preprocessed dataset, before the holdout split. Using separate partitioners

per model avoids loop-conflict issues in KNIME.

Clustering. K-Means is applied to 10 features (`sqft_living`, `grade`, `lat`, `long`, `price`, `bathrooms`, `view`, `waterfront`, `house_age`, `renovated`) after Z-score normalisation on the full dataset. $K \in \{2, 3, 4, 5\}$ is evaluated by the Silhouette Coefficient [5].

Results

Regression performance

Table 2 reports holdout and CV results. RF achieves the highest R^2 on both evaluations, confirming the advantage of bagging ensembles. GBT ranks second; its stronger CV relative to holdout suggests it generalises well but is slightly more sensitive to training variance. LR is a competitive and interpretable third. DT performs worst in CV (RMSE=0.327) despite a moderate holdout score, indicating sensitivity to the specific train/test split. LR+PCA ($R^2=0.654$) falls below full LR (0.663), quantifying the information cost of reducing 14 features to 12 PCs — a small but measurable loss.

Table 2: Holdout test set and 5-fold CV results (target: `log_price`).

Model	Test R^2	Test RMSE	CV Avg MSE	CV RMSE
Random Forest	0.703	0.292	0.0820	0.286
Gradient Boosted Trees	0.696	0.295	0.0874	0.296
Linear Regression	0.663	0.311	0.0949	0.308
LR + PCA (12 dims)	0.654	0.315	—	—
Decision Tree	0.620	0.330	0.1070	0.327

Residual analysis

Figure 2 shows RF residuals on the test set. The histogram is bell-shaped and centred at zero (mean signed difference = -0.00025), indicating that predictions are unbiased. The scatter plot of residuals versus predicted values shows no systematic pattern, confirming homoscedasticity. Both results support the validity of the regression assumptions.

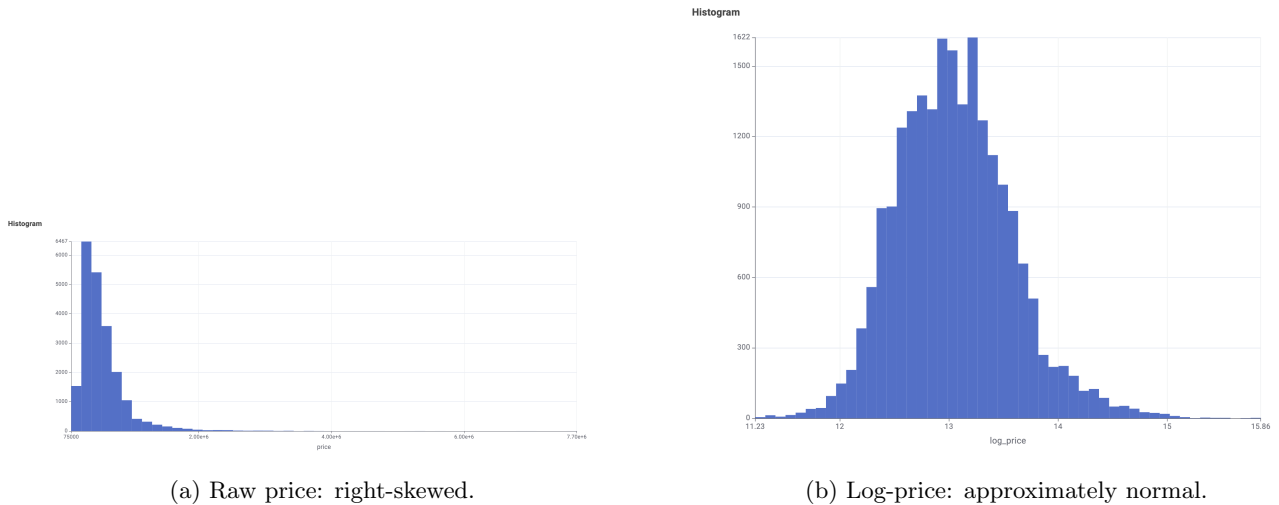


Figure 1: Log transformation normalises the target distribution, motivating its use as the regression target.

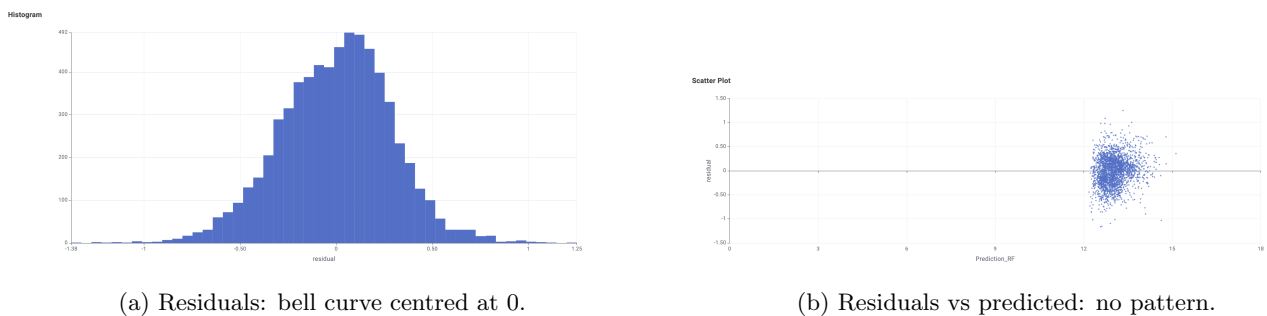


Figure 2: Residual analysis for Random Forest (holdout test set). Both plots confirm unbiased, homoscedastic errors.

Market segmentation

Table 3 shows Silhouette scores decreasing monotonically with K , selecting $K = 2$ as optimal. Figure 3 shows the geographic distribution. Two segments emerge: a *standard market* (Cluster 0: 13,961 houses, 65%, below-average price and living space) and a *premium market* (Cluster 1: 7,651 houses, 35%, price +0.70 SD, living space +0.94 SD, grade +0.95 SD, higher waterfront proportion). The geographic separation visible in Figure 3 suggests that location is a key driver of cluster membership, consistent with `lat` and `long` being included among the clustering features.

Table 3: Silhouette scores.

K	Silhouette
2	0.248
3	0.217
4	0.199
5	0.182

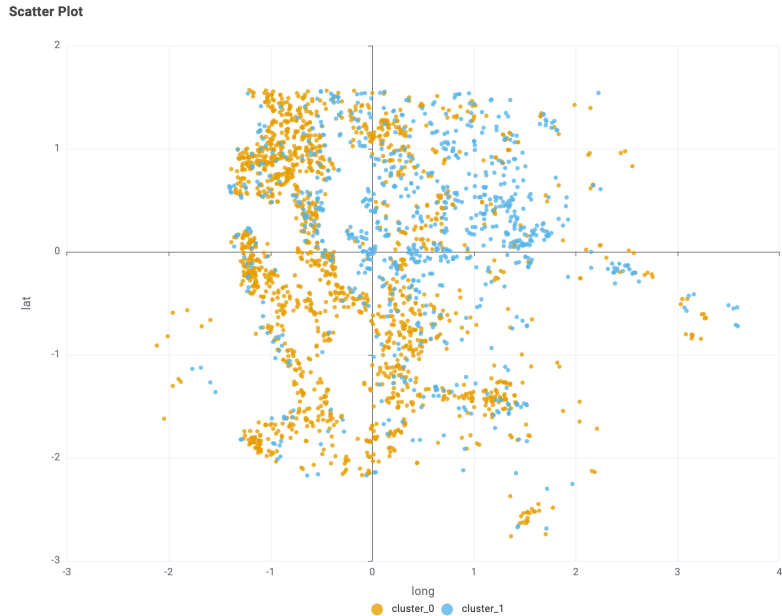


Figure 3: K-Means ($K = 2$) on normalised lat/long. Orange = standard market (Cluster 0); blue = premium market (Cluster 1). Geographic separation is visible.

Feature importance

RF attribute statistics (number of splits at tree level 0) confirm that `bathrooms` (20), `sqft_living15` (19), `sqft_living` (16), `grade` (13), and `floors` (11) are used most at the root of trees — consistent with the correlation structure in Table 1. Features such as `condition`, `waterfront`, and `renovated` have zero root splits, indicating that while they carry some signal, they are less discriminative at the top level of the decision trees.

Conclusions

Random Forest is the best model for predicting King County house prices ($R^2=0.703$), a result that is stable across the holdout evaluation and 5-fold cross-validation. GBT is a competitive alternative; LR provides a useful interpretable baseline. PCA with 12 components (95% variance) achieves $R^2=0.654$, a modest reduction from LR at full dimensionality (0.663) — demonstrating that the 14 features contain limited redundancy beyond what 12 PCs can capture. K-Means clustering reveals a premium/standard market split that aligns with geographic location and property quality.

The main limitation is temporal and geographic scope: the data covers a single year and region, and does not include macroeconomic variables. Future work could incorporate spatial autocorrelation models and time-series components.

Reproducibility. The full KNIME workflow, dataset, and plots are available at: <https://github.com/aminentezari/Machine-learning-house-price>



Scan to access the repository

References

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